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● PS D:\ODD SEM\DSA3\Project\SmartShopping> javac *.java
○ PS D:\ODD SEM\DSA3\Project\SmartShopping> java Main products.csv
=====
Dictionary-Based Fuzzy Search & Spell Correction
Engine for Smart Shopping
=====
Catalog loaded: 122 items across 10 categories

1. Search for an item (with spell correction)
2. Browse all items
3. Browse items by category
4. Autocomplete (type a prefix)
5. Keyword / substring search
6. Trending searches
7. Search by price range (Binary Search)
8. Top-N cheapest / most expensive (Heap)
9. Shopping cart
10. View catalog structure (Trie)
11. Exit
Choose an option: █

```

```

Choose an option: 1
Enter item name: iphn
Did you mean: "iphone"?

Item found:
Name       : Iphone
Category   : Electronics
Price      : Rs. 79999.00
Description : Iphone - a reliable electronics product for everyday use.

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```

Choose an option: 2

--- All Items (122) ---

Airconditioner	HomeAppliances	Rs.34999.00
Basketball	Sports	Rs.999.00
Backpack	Accessories	Rs.1299.00
Badminton	Sports	Rs.799.00
Bed	Furniture	Rs.12999.00
Bedsheet	Furniture	Rs.999.00
Belt	Accessories	Rs.499.00
Bicycle	Sports	Rs.8999.00
Blazer	Apparel	Rs.2999.00
Blender	HomeAppliances	Rs.1999.00
Bluetooth	Accessories	Rs.1299.00
Boots	Footwear	Rs.2499.00
Bookshelf	Furniture	Rs.3999.00
Cap	Accessories	Rs.299.00
Cable	Accessories	Rs.299.00
Camera	Electronics	Rs.24999.00
Crayons	Stationery	Rs.99.00
Cricketbat	Sports	Rs.1499.00
Curtain	Furniture	Rs.799.00
Charger	Accessories	Rs.499.00
Chair	Furniture	Rs.2499.00
Choppingboard	Kitchen	Rs.399.00
Cyclehelmet	Sports	Rs.999.00
Conditioner	Beauty	Rs.349.00
Dress	Apparel	Rs.1599.00

Choose an option: 3

Available categories:

1. Accessories
2. Apparel
3. Beauty
4. Electronics
5. Footwear
6. Furniture
7. HomeAppliances
8. Kitchen
9. Sports
10. Stationery

Type a category name: kitchen

--- Kitchen (10 items) ---

Toaster	Kitchen	Rs.1299.00
Inductioncooktop	Kitchen	Rs.2299.00
Pressurecooker	Kitchen	Rs.1899.00
Frypan	Kitchen	Rs.899.00
Saucepan	Kitchen	Rs.799.00
Knife	Kitchen	Rs.299.00
Choppingboard	Kitchen	Rs.399.00
Waterbottle	Kitchen	Rs.299.00
Lunchbox	Kitchen	Rs.499.00
Thermosflask	Kitchen	Rs.699.00

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11. Exit

Choose an option: █

Choose an option: 4

Start typing an item name: sha

Suggestions:

- Sharpener	Stationery	Rs.15.00
- Shampoo	Beauty	Rs.299.00

1. Search for an item (with spell correction)
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3. Browse items by category
4. Autocomplete (type a prefix)
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6. Trending searches
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9. Shopping cart
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11. Exit

Choose an option: █

```
Choose an option: 5
Enter keyword: book
Items containing "book":
- Bookshelf      | Furniture      | Rs.3999.00
- Notebook       | Electronics    | Rs.49999.00
- Notebook2      | Stationery     | Rs.50.00

1. Search for an item (with spell correction)
2. Browse all items
3. Browse items by category
4. Autocomplete (type a prefix)
5. Keyword / substring search
6. Trending searches
7. Search by price range (Binary Search)
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11. Exit
Choose an option: █
```

```
Choose an option: 6

--- Trending Searches ---
1. bskball (1 searches)
2. iphn (1 searches)
```

Choose an option: 7

Minimum price: 50

Maximum price: 2000

--- Items between Rs.50.0 and Rs.2000.0 ---

Notebook2	Stationery	Rs.50.00
Scissors	Stationery	Rs.79.00
Toothpaste	Beauty	Rs.89.00
Crayons	Stationery	Rs.99.00
Stapler	Stationery	Rs.99.00
Soap	Beauty	Rs.99.00
Flipflops	Footwear	Rs.199.00
Skippingrope	Sports	Rs.199.00
Facewash	Beauty	Rs.249.00
Cap	Accessories	Rs.299.00
Cable	Accessories	Rs.299.00
Deodorant	Beauty	Rs.299.00
Knife	Kitchen	Rs.299.00
Shampoo	Beauty	Rs.299.00
Slippers	Footwear	Rs.299.00
Waterbottle	Kitchen	Rs.299.00
Conditioner	Beauty	Rs.349.00
Choppingboard	Kitchen	Rs.399.00
Moisturizer	Beauty	Rs.399.00
Scarf	Accessories	Rs.399.00
Tshirt	Apparel	Rs.399.00
Sunscreen	Beauty	Rs.449.00
Belt	Accessories	Rs.499.00
Charger	Accessories	Rs.499.00
Lunchbox	Kitchen	Rs.499.00
Pendrive	Electronics	Rs.499.00
Pillow	Furniture	Rs.499.00
Sandals	Footwear	Rs.499.00
Umbrella	Accessories	Rs.499.00
Lamp	HomeAppliances	Rs.599.00

Choose an option: 8

Available categories:

- Accessories
- Apparel
- Beauty
- Electronics
- Footwear
- Furniture
- HomeAppliances
- Kitchen
- Sports
- Stationery

Category (or 'all' for entire catalog): Apparel

Cheapest or most expensive? (cheap/expensive): cheapest

How many items (N)? 3

--- Top 3 (cheapest) ---

Tshirt	Apparel	Rs.399.00
Shorts	Apparel	Rs.599.00
shirt	Apparel	Rs.699.00

Choose an option: 9

--- Shopping Cart ---

1. Add item
2. View cart & total
3. Remove item
4. Back to main menu

Choose an option: 1

Item name: basketball

Quantity: 2

Added 2 x basketball to cart.

--- Shopping Cart ---

1. Add item
2. View cart & total
3. Remove item
4. Back to main menu

Choose an option: 1

Item name: bookshelf

Quantity: 1

Added 1 x bookshelf to cart.

--- Shopping Cart ---

1. Add item
2. View cart & total
3. Remove item
4. Back to main menu

Choose an option: 2

--- Your Cart ---

Basketball	x2	Rs.999.00	Subtotal: Rs.1998.00
Bookshelf	x1	Rs.3999.00	Subtotal: Rs.3999.00

Total: Rs.5997.00

--- Shopping Cart ---

1. Add item
2. View cart & total
3. Remove item
4. Back to main menu

Choose an option: 3

Item name to remove: bookshelf

Removed.

--- Shopping Cart ---

1. Add item
2. View cart & total
3. Remove item
4. Back to main menu

Choose an option: 2

--- Your Cart ---

Basketball	x2	Rs.999.00	Subtotal: Rs.1998.00
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Total: Rs.1998.00

--- Shopping Cart ---

1. Add item
2. View cart & total
3. Remove item
4. Back to main menu

Choose an option: 4

Choose an option: 10

--- Catalog Structure (Trie) ---

(root)

```
|-- a
|   |-- i
|   |   |-- r
|   |       |-- c
|   |           |-- o
|   |               |-- n
|   |                   |-- d
|   |                       |-- i
|   |                           |-- t
|   |                               |-- i
|   |                                   |-- o
|   |                                       |-- n
|   |                                           |-- e
|   |                                               |-- r [word end]
|-- b
|   |-- a
|   |   |-- s
|   |       |-- k
|   |           |-- e
|   |               |-- t
|   |                   |-- b
|   |                       |-- a
|   |                           |-- l
|   |                               |-- l [word end]
|   |-- c
|   |   |-- k
|   |       |-- p
|   |           |-- a
|   |               |-- c
```

Choose an option: 11

Thank you for shopping with us!

PS D:\ODD SEM\DSA3\Project\SmartShopping> █